

STAA57 Draft Notebook

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Description of the Data

Description of the variables and the data
answer: will be given in the repo

Variable	Explanation
Var1	Description

Background of Data

A little background about the data (who collected it in what context)
answer: will be given in the repo

The data used in this report was collected and published by the Government of Ontario through the open data portal. It includes datasets related to wages by education level, labour force estimates by educational attainment, and employment characteristics of students. These datasets are designed to provide insight into economic and workforce trends across the province and are regularly updated to support public research and policy analysis.

For this project, we focus on data that captures the relationship between post-secondary education level and career outcomes. Key variables include average wages, employment and unemployment rates, and labour force participation across different levels of educational attainment (e.g., less than high school, high school diploma, college, and university). These variables include both quantitative data (such as wages and rates) and categorical data (such as education level), making them well-suited for analysis using statistical and data science methods.

By working with this real-world dataset, we aim to identify patterns and relationships between education and career success, compare outcomes across different education levels, and evaluate how strongly education level is associated with economic opportunities in Ontario.

This data is publicly available through Ontario's open data catalogue and is licensed under the Open Government Licence Ontario.

Dataset 1

Dataset 2

Dataset 3

Dataset 4

Research Question

What is the over all research question that you are trying to answer in your report

How does post-secondary education level affect career outcomes and measures of career success in Ontario?

We explore this through three key questions:

1. How does education level affect average wages in Ontario?
2. How does education level influence employment and unemployment rates?
3. How does education level affect participation in the labour force (i.e., likelihood of being employed or actively seeking work)?

Additionally, we aim to develop a predictive model to estimate career outcomes (such as wages and employment status) based on education level.

Analysis portion

Few tables (appropriately labeled), Few graphs (appropriately labeled)

Use of confidence interval and test of hypothesis

Use of bootstrapping, Use of regression analysis (linear or nonlinear regression [week-9] or logistic regression[week-10] or decision tree/random forest [Week-11]). Interpretation of the regression parameters when applicable.

Use of cross validation

Final Summary

A final summary of your findings

In conclusion:

Appendix

An appendix which shows all your codes

answer: will be given in the repo

NOTE: 3 marks will be kept aside for the presentation of your report (e.g. proper title, subtitle, graph size, graph placing, line spacing, proper annotation of your code etc.)